

global





# Stochastic Storm Surge Modeling using ADCIRC+SWAN through Parallel Computing

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# Outline





# Introduction



- ▶ Typhoon Haiyan brought destructive storm surge in Tacloban City on Nov 08 2013



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- ▶ Storm surge predictions from different agencies are not completely consistent
- ▶ No CFD software for stochastic storm surge
- ▶ We need better and more realistic model!



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- ▶ Compute fast enough and give real-time predictions



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- ▶ Include stochastic process in the governing equations of ADCIRC+SWAN model



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- ▶ Compute fast enough and give real-time predictions
- ▶ Include stochastic process in the governing equations of ADCIRC+SWAN model
- ▶ Investigate parameter sensitivity



## Related Studies



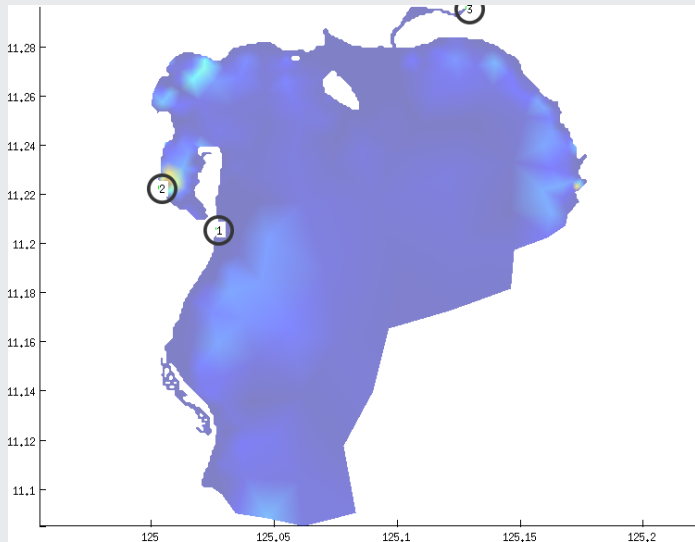
Can be done using deterministic models

- ▶ ADCIRC+SWAN
- ▶ Delft3D-FLOW+WAVE
- ▶ SLOSH
- ▶ JMA

Only ADCIRC+SWAN can compute in [parallel](#)

Can also use statistical analysis based from past events

- ▶ Highly developed program for solving fluid motion equations in 2D/3D
- ▶ Finite-element in time and finite-difference in time
- ▶ Uses shallow water equations and continuity equation to obtain water velocity and elevation
- ▶ Grid flexibility, efficiency and accuracy well documented





# Storm Surge Prediction



- ▶ Gaussian noise is added to the force vector to simulate additive random forcing, requires code modification

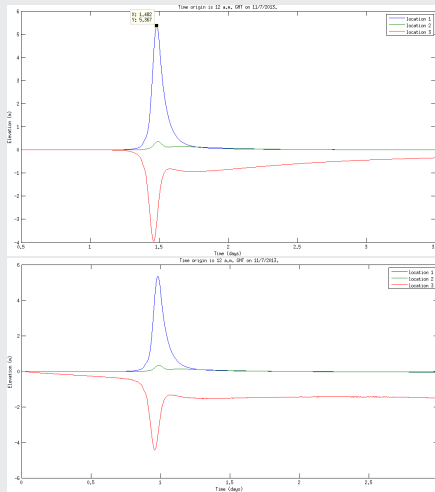


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- ▶ Random forces exerted by some debris



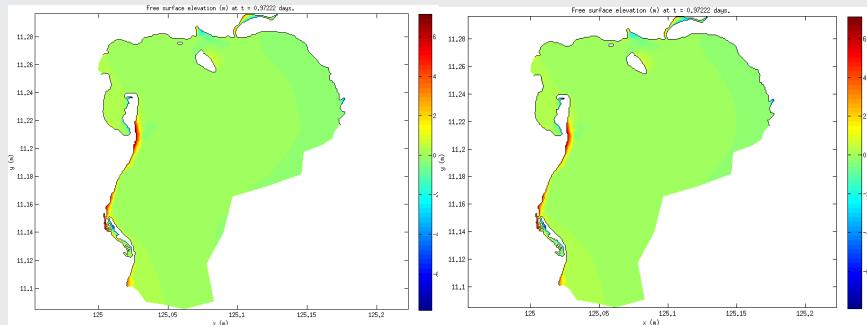
- ▶ Gaussian noise is added to the force vector to simulate additive random forcing, requires code modification
- ▶ Random forces exerted by some debris
- ▶ Zero mean and variance equals  $10^{-4}$

# Without vs with noise



Model is very sensitive to noise, might lead to unstable computation

# Without vs with noise



Contours still show coherent structures despite noise



# Sensitivity Analysis



- ▶ If the water total depth is larger than  $H_{min}$ , the node remains "wet" and included in the rest of calculations.
- ▶ Else, the node is "dry" and removed from the calculations.

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- ▶ This criterion almost becomes a height restriction if the adj node's free surface circulation is sufficiently larger than its own.

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- ▶ Note that  $\tau \propto \tau_0$
- ▶  $\tau_i$  increases in shallower, near-shore waters and directly affects the speed at which waves inundate or recede.

- ▶ The parameter  $G$  is the numerical parameter that controls the balance between the wave continuity equation and the primitive continuity equation.

$$\frac{\partial \zeta}{\partial t} + G\zeta - \nabla \cdot M = 0$$

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- ▶ As  $G \rightarrow 0$ : pure wave equation,  $G \rightarrow \infty$ : pure primitive equation.
- ▶ If  $G$  is set too high, solution has spurious oscillations that prevent the model from capturing behavior of waves.



Parameters are adjusted one by one from the set of default values



- ▶ Lower  $H_{min}$ , higher surge
- ▶ Higher  $G$ , more oscillations, unstable. Lower  $G$ , higher surge height; water subsides slower
- ▶ Lower  $\tau_0$ , higher surge height; water subsides slower



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- ▶ After Gaussian noise, we also consider colored noise and Levy process
- ▶ We are only concerned with the **numerical** solution of the SPDE

# Conclusion





R. A. Luettich, Jr., et. al.

*ADCIRC: An Advanced Three-Dimensional Circulation Model, Theory and Methodology.*

US Army Engineer Waterways Experiment Station, Vicksburg, Mississippi, 1992.



J. C. Dietrich, et. al.

*Assessment of ADCIRC's Wetting and Drying Algorithm.*

Institute of Marine Sciences, University of North Carolina, 2005.

Thank you for your attention!

